

LEDGER · 2026-09-04 · AGROCHEMICALS / NANOTECHNOLOGY

# Nitrogen-Doped Carbon Nanodots (N-CDs) via Atmospheric Microwave Pyrolysis

PRODUCT



A product dossier of The Absence Audit. The current audit flags this concept — reserved findings — and the full disclosure ships inside the dossier.

---

**Incumbent benchmark**

Mancozeb 75 WP

---

**Entry application**

Field-scale foliar application targeting Late Blight (*Phytophthora infestans*) in commercial potato and tomato farming

---

|                         |                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| Measured advantage      | 40 µg/mL MIC for N-CD active vs >500 µg/mL for Mancozeb 75 WP                                                      |
| Time to first revenue   | 36 months                                                                                                          |
| Gross margin at parity  | 88.4%                                                                                                              |
| Startup CapEx           | \$14,000                                                                                                           |
| Payback from first sale | 0.7 months                                                                                                         |
| Capital productivity    | 61.71x                                                                                                             |
| Regulatory risk         | Moderate — qualification costs reported (\$2,000,000 estimated)                                                    |
| Market absence          | Verified — the search is reported in the dossier: channels searched, companies searched, closest substitutes named |

#### BUY THE FULL DOSSIER

Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

**[Buy the full dossier — \\$299.00 →](#)**

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

Published for research and educational purposes. Nothing here is investment, legal, regulatory or engineering advice, and no commercial outcome is promised or implied. Concepts are drawn from published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.

© 2026 The Absence Audit. All rights reserved. · [Terms of Use](#)